

Eunhan Ka

POSTDOCTORAL RESEARCHER · LYLES SCHOOL OF CIVIL AND CONSTRUCTION ENGINEERING, PURDUE UNIVERSITY
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Education

Purdue University

United States

PH.D., LYLES SCHOOL OF CIVIL AND CONSTRUCTION ENGINEERING

Dec. 2025

- **Dissertation: Physics-Informed Neural Networks for Secure Connected and Autonomous Traffic Modeling**
- Committee: Satish V. Ukkusuri (Chair), Ludovic Leclercq, Yiheng Feng, Z. Berkay Celik

Seoul National University

South Korea

M.S., DEPARTMENT OF CIVIL AND ENVIRONMENTAL ENGINEERING

Feb. 2018

Seoul National University

South Korea

B.S., DEPARTMENT OF CIVIL AND ENVIRONMENTAL ENGINEERING

Feb. 2016

- Completed the ABEEK-accredited Civil and Environmental Engineering program (Engineering Accreditation Commission; Washington Accord).

Research Interests

Modeling of network traffic dynamics, resilience, and large-scale state estimation, using physics-informed and data-driven methods that integrate behavioral models, game theory, and optimization across traffic engineering and machine learning.

- **Themes:** Network Traffic Dynamics, Large-Scale Traffic State Estimation, Network Resilience and Reliability, Connected and Automated Mobility, Human Behavior and Safety, Transportation Cybersecurity.
- **Methods:** Physics-Informed Learning, Graph & Network Learning, Game-Theoretic Modeling, Symbolic Learning and Scientific Discovery, Optimization and Simulation, Spatiotemporal Data Analytics

Selected Publications & Awards

Selected Publications

1. **Ka, E.**, Xue, J., Leclercq, L., and Ukkusuri, S. V. (2024). A Physics-Informed Machine Learning for Estimating Traffic State with a Generalized Bathtub Model in Large-scale Urban Networks. *Transportation Research Part C: Emerging Technologies*, 164, 104661.
2. **Ka, E.** and Ukkusuri, S. V. (2025). Route Guidance Attacks in Cyber Transportation Networks: A User-Centered Study of Behavioral Sensitivity. *Transportation Research Part F: Traffic Psychology and Behaviour*, 115, 103354.
3. Xue, J., **Ka, E.**, Feng, Y., and Ukkusuri, S. V. (2024). Network Macroscopic Fundamental Diagram-Informed Graph Learning for Traffic State Imputation. *Transportation Research Part B: Methodological*, 189, 102996.
4. **Ka, E.**, Sharma, S., and Ukkusuri, S. V. (2022). Leveraging Location-Based Data for Assessing Network-Level Traffic Impact of Lane Management: A Case Study of Alex Fraser Bridge. *Journal of Transportation Engineering, Part A: Systems*, 148(12), 04022105. [2022 Editor's Choice Collections]

Selected Awards

1. **2025 ITE Great Lakes District Student Paper Award** (First Place in regional student paper competition), *Great Lakes District of ITE (GLITE)*. Jun. 2025
2. **Honorable Mention, 2025 Clifford Spiegelman Student Paper Competition**, *The Transportation Statistics Interest Group of the American Statistical Association*. Jan. 2025
3. **Google Cloud Research Credits**, *Google Cloud*. Jun. 2026
4. **Kinnier Graduate Scholarship & STV Civil Engineering Graduate Assistantship Endowment**, *Lyles School of Civil and Construction Engineering, Purdue University*. Oct. 2024

Research Experience

Postdoctoral Researcher

Jan. 2026 - Present

PURDUE UNIVERSITY, UNITED STATES

- Develop physics-informed and data-assimilation methods for **transportation network state estimation, reliability assessment, and decision support** under sparse and disrupted observations.
- Build a **Resilient Mobility Digital Twin** to estimate network conditions, quantify uncertainty, and evaluate disruption propagation and recovery under climate, infrastructure, and cyber scenarios.
- Develop **behavior-aware models of connected-mobility disruptions** (e.g., route-guidance manipulation in navigation services) to evaluate network performance, operational risk, and mitigation strategies.
- Build **scalable simulation and graph-based analytical tools** for reproducible transportation-system evaluation and proposal development.

Graduate Research Assistant

Aug. 2020 - Dec. 2025

PURDUE UNIVERSITY, UNITED STATES

- Developed physics-informed and graph-based methods for **large-scale traffic-state estimation, network operations, and resilience under sparse or disrupted data**, resulting in publications in **Transportation Research Part B, Part C, and Part F**. (Sponsor: USDOT University Transportation Center-TraCR)
- Conducted a statewide workforce needs assessment to identify skill gaps and designed training strategies to improve engagement and performance (presented at **Purdue Road School, Mar. 2025**). (Sponsor: INDOT)
- Analyzed network-level impacts of a movable median barrier on the Alex Fraser Bridge using mobile location data (Journal of Transportation Engineering, Part A, 2022; **Editor's Choice**). (Sponsor: Lindsay Corporation)
- Led proposal development and served as the **primary technical lead throughout project execution**, coordinating analytical work, project milestones, and deliverable preparation under the supervision of Dr. Satish V. Ukkusuri (Principal Investigator). **Selected Sponsored Projects:**
 1. Attracting and Retaining the Transportation Workforce and At Risk Targeted Areas (Sponsor: INDOT; total project award: \$225,000; Oct. 2024–Sep. 2026).
 2. A Multi-Resolution Simulation Platform for Transportation System Security Testing and Evaluation (Sponsor: USDOT UTC-TraCR; total project budget: \$340,000 including cost share; Jan.–Dec. 2024).
 3. Training Gap Analysis for INDOT Workforce (Sponsor: INDOT; total project award: \$213,500; Oct. 2023–Oct. 2025).

Researcher

Mar. 2018 - Aug. 2020

SEOUL NATIONAL UNIVERSITY, SOUTH KOREA

- Designed cluster-based routes for demand-responsive transport using multi-capacity vehicles to serve passengers with disabilities (presented at **99th TRBAM**). (Sponsor: T-Money Welfare Foundation)
- Modeled lane-changing on freeways and gap-acceptance behavior at roundabouts using real-world data and virtual reality simulations (published in **Journal of Advanced Transportation, 2018**). (Sponsor: Ministry of Land, Infrastructure and Transport, South Korea)

Graduate Research Assistant

Mar. 2016 - Feb. 2018

SEOUL NATIONAL UNIVERSITY, SOUTH KOREA

- Designed a real-time vehicle relocation strategy and an event-based simulation for one-way car-sharing services (presented at **multiple conferences, 2016-2018**). (Sponsor: National Research Foundation of Korea)
- Developed a methodology to evaluate driver behavior using surrogate safety measures (published in **Journal of Advanced Transportation, 2020**). (Sponsor: Ministry of Land, Infrastructure and Transport, South Korea)

Teaching & Mentoring Experience

Research Mentor

Fall 2024 - Present

LYLES SCHOOL OF CIVIL AND CONSTRUCTION ENGINEERING, PURDUE UNIVERSITY

- **Graduate research mentoring (Jun. 2026–Present):** Mentor two Ph.D. students in transportation systems research; each student submitted a paper to the 2027 Transportation Research Board Annual Meeting.
- **Undergraduate research mentoring (Fall 2024–Spring 2025):** Mentored two undergraduate researchers in transportation cybersecurity; their work resulted in a poster presentation at the 2025 CERIAS Cybersecurity Symposium.

Teaching Assistant

Spring 2021 - Fall 2024

LYLES SCHOOL OF CIVIL AND CONSTRUCTION ENGINEERING, PURDUE UNIVERSITY

- CE 661: Algorithms in Transportation (*Graduate; co-developed new course curriculum*) Fall 2024
- CE 597: Network Models for Connected and Autonomous Vehicles (Graduate) Fall 2021 - Fall 2023
- CE 597: Smart Logistics (Graduate) Fall 2021, Fall 2024
- CE 597: Data Science for Smart Cities (Graduate) (*Developed online course on edX*) Spring 2021, Fall 2023

Lecturer

Fall 2019

DEPT. OF CIVIL AND ENVIRONMENTAL ENGINEERING, SEOUL NATIONAL UNIVERSITY

- Introduction to Civil and Environmental Engineering (Undergraduate)

Teaching Assistant

Fall 2016 - Fall 2017

DEPT. OF CIVIL AND ENVIRONMENTAL ENGINEERING, SEOUL NATIONAL UNIVERSITY

- Sustainable Transportation Systems (Graduate) Fall 2017
- Advanced Transportation Operation (Graduate) Spring 2017
- Leadership for Civil Engineers (Undergraduate) Spring 2017
- Introduction to Transportation Engineering (Undergraduate) Fall 2016

Publications

MANUSCRIPTS UNDER REVIEW AND REVISION

1. **Ka, E.**, Leclercq, L., and Ukkusuri, S. V. (Under Major Revision). Adaptive Domain Decomposition Physics-Informed Neural Networks for Traffic State Estimation with Sparse Sensor Data. *Transportation Research Part C: Emerging Technologies*.
2. **Ka, E.**, Mittal, S., Ali, A., Liu, L., and Ukkusuri, S. V. (Under Review). Perceived regulatory calibration and cybersecurity workforce capacity in U.S. transportation: Exploratory survey evidence. *Transport Policy*.
3. **Ka, E.** and Ukkusuri, S. V. (Under Review). Day-to-Day Traffic Network Modeling under Route-Guidance Misinformation: Endogenous Trust and Resilience in CAV Environments. *IEEE Transactions on Intelligent Transportation Systems*.
4. **Ka, E.**, Yeke, D., Celik, Z. B., and Ukkusuri, S. V. (Under Review). Fake Rush Hour: Exploring Bounded Rational Route Guidance Attacks on Urban Navigation Systems. *ACM Journal on Autonomous Transportation Systems*.
5. Yang, Z., **Ka, E.**, Cai, Z., and Ukkusuri, S. V. (Under Review). Context-Matched Scenario Families with Controlled Criticality for Autonomous Driving Testing. *Transportation Research Record*.
6. **Ka, E.**, Li, H., and Ukkusuri, S. V. (Under Review). Finite-Grid Used-Path Regret Certificates for Restricted Static Traffic Assignment. *Transportation Research Record*.

REFEREED JOURNAL PUBLICATIONS

1. Ali, A., Liu, L., Ojeme, B., Anadozie, C., Agyemang, F., Ukkusuri, S., **Ka, E.**, and Mittal, S. (Accepted). Strengthening the Transportation Cybersecurity Workforce: A Mixed-Methods Analysis of Workforce Development, Organizational Preparedness and Cross-Sector Collaboration. *Future Transportation*.
2. **Ka, E.**, Mittal, S., and Ukkusuri, S. V. (Accepted). Career-Stage-Specific Training Priorities for State DOT Workforce Development: An Integrated Regression and Importance-Performance Analysis. *Transportation Research Record: Journal of the Transportation Research Board*.
3. **Ka, E.** and Ukkusuri, S. V. (2026). Resilience of Traffic Networks to Route Guidance Attacks: The Role of Driver Behaviour Heterogeneity. *Transportmetrica B: Transport Dynamics*, 14, 1.
4. **Ka, E.** and Ukkusuri, S. V. (2025). Route Guidance Attacks in Cyber Transportation Networks: A User-Centered Study of Behavioral Sensitivity. *Transportation Research Part F: Traffic Psychology and Behaviour*, 115, 103354.
5. Ukkusuri, S. V.*, Hamim, O. F.*, Lei, Z. *, **Ka, E. ***, Salek, M. S., Chowdhury, M., Amini, M. H., Cardenas, A., and Thuraisingham B. (2025). Cybersecurity for Next-Generation Road Transportation: A Review. *ACM Journal on Autonomous Transportation Systems*. (*Contributed equally to this research).
6. Xue, J., **Ka, E.**, Feng, Y., and Ukkusuri, S. V. (2024). Network Macroscopic Fundamental Diagram-Informed Graph Learning for Traffic State Imputation. *Transportation Research Part B: Methodological*, 189, 102996.

7. **Ka, E.**, Xue, J., Leclercq, L., and Ukkusuri, S. V. (2024). A Physics-Informed Machine Learning for Estimating Traffic State with a Generalized Bathtub Model in Large-scale Urban Networks. *Transportation Research Part C: Emerging Technologies*, 164, 104661.
8. **Ka, E.**, Sharma, S., and Ukkusuri, S. V. (2022). Leveraging Location-Based Data for Assessing Network-Level Traffic Impact of Lane Management: A Case Study of Alex Fraser Bridge. *Journal of Transportation Engineering, Part A: Systems*, 148(12), 04022105. **[2022 Editor's Choice Collections]**
9. **Ka, E.**, Kim, D. G., Hong, J., and Lee, C. (2020). Implementing Surrogate Safety Measures in Driving Simulator and Evaluating the Safety Effects of Simulator-Based Training on Risky Driving Behaviors. *Journal of Advanced Transportation*, 2020(1), 7525721.
10. Lee, D., Hwang, S., **Ka, E.**, and Lee, C. (2018). Evaluation of the Rain Effects on Gap Acceptance Behavior at Roundabouts by a Logit Model. *Journal of Advanced Transportation*, 2018(1), 2726732.

CONFERENCE PAPERS AND PRESENTATIONS

1. Ortiz Barbosa, D., Abdisho, E. B., Chang, O., Burbano, L., Lei, Z., **Ka, E.**, Ukkusuri, S. V., Gilpin, L. H., and Cardenas, A. A. (2026). Beyond Attacker-Defender Ratios in Counter-Swarm Defense. *Dynamic Data Driven Applications Systems (DDDAS) 6th International Conference*. New Brunswick, NJ, United States
2. **Ka, E.**, Yeke, D., Celik, Z. B., and Ukkusuri, S. V. (2026). MIRAGE: Detecting Fake Emergency Electronic Brake Light Attacks in V2X Networks via Event-Gated Behavioral Analysis. *4th USENIX Symposium on Vehicle Security and Privacy (VehicleSec'26)*. Baltimore, MD, United States
3. **Ka, E.** and Ukkusuri, S. V. (2025). Analytical Framework for Network-Level Traffic Flow under Route Guidance Attacks: An Extension of the Generalized Bathtub Model. *104th Annual Meeting of the Transportation Research Board*, Washington, D.C., United States.
4. **Ka, E.** and Ukkusuri, S. V. (2025). Comparative Analysis of Sampling Methods in Physics-Informed Neural Networks for Traffic State Estimation in Large-Scale Road Networks. *104th Annual Meeting of the Transportation Research Board*, Washington, D.C., United States.
5. **Ka, E.**, and Ukkusuri, S. V. (2024). Impact of Cyber Attacks on Traffic State Estimation for Connected and Autonomous Vehicles Systems. *The Inaugural USDOT Future of Transportation (FoT) Summit*, Washington, D.C., United States.
6. Xue, J., **Ka, E.**, and Ukkusuri, S. V. (2024). Network Macroscopic Fundamental Diagram-Informed Graph Learning for Traffic State Imputation. *ISTTT25: 25th International Symposium on Transportation and Traffic Theory*, Ann Arbor, MI, United States.
7. **Ka, E.**, Xue, J., and Ukkusuri, S. V. (2024). PIDL-PedFlow: A Physics-Informed Deep Learning Approach for Macroscopic Continuum Pedestrian Flow Modelling. *103rd Annual Meeting of the Transportation Research Board*, Washington, D.C., United States.
8. Xue, J., **Ka, E.**, Mondal, W. U., and Ukkusuri, S. V. (2024). Generating Network-Level Dynamic Traffic Equations Using Symbolic Regression. *103rd Annual Meeting of the Transportation Research Board*, Washington, D.C., United States.
9. Sharma, S., and **Ka, E.** (2024). Leveraging Location-Based Data for Assessing Network Level Traffic Impact of Lane Management: A Case Study of Alex Fraser Bridge. *103rd Annual Meeting of the Transportation Research Board*, Washington, D.C., United States.
10. **Ka, E.**, and Ukkusuri, S. V. (2023). Dynamic Routing Games for Connected and Autonomous Vehicles with Traffic Congestion: A Mean Field Game Approach. *2023 INFORMS Annual Meeting*, Phoenix, AZ, United States.
11. **Ka, E.**, Ka, D., Jung, Y., and Lee, C. (2020). A Cluster-Based Route Design of Multi-Capacity Vehicle in Large-Scale Demand Responsive Transport Service for the Disabled. *99th Annual Meeting of the Transportation Research Board*, Washington, D.C., United States.
12. **Ka, E.**, Hong, D., Na, Y., and Lee, C. (2018). Analysis of Status in DRT Service for the Disabled in Seoul and Comparison Domestic and Foreign Cases. *International Conference for Road Engineers*, Jeju, South Korea.
13. Hong, D., **Ka, E.**, Ha, S., and Lee, C. (2018). Selection of Appropriate Hyperparameter for Waiting Time Prediction Model for Demand Responsive Transport for the Disabled in Seoul Using Long Short-Term Memory (LSTM) Network. *78th Korean Society of Transportation Conference*, Wonju, South Korea. **[Outstanding Paper Award]**.
14. **Ka, E.**, Kim, S., Hong, J., and Lee, C. (2017). A Preliminary Study of Comparison with Lane Changing Model Parameters in Merging Area between Normal and Raining Conditions. *12th International Conference of Eastern Asia's Society for Transportation Studies (EASTS)*, Ho Chi Minh, Vietnam.

15. **Ka, E.**, Kim, S., Woo, D., and Lee, C. (2016). An Estimation of Critical Gap for Gap Acceptance Model Applied to Lane Change of Surrounding Vehicles in Driving Simulator. *2016 Fall Korea Institute of ITS Conference*, Jeju, South Korea.
16. Lee, S., Lee, H., Lee, J., **Ka, E.**, and Lee, C. (2016). Analysis of Traffic Flow Impacts of Highway Traffic Accidents Using Survival Analysis. *74th Korean Society of Transportation Conference*, Jeju, South Korea. **[Outstanding Paper Award]**.

BOOK CHAPTERS AND TECHNICAL REPORTS

1. **Ka, E.** and Ukkusuri, S. V. (2026). Misdirected Guidance and Urban Traffic Network Resilience. In *Advances in Transportation Cybersecurity and Resiliency*. World Scientific.
2. Hamim, O. F., Lei, Z., **Ka, E.**, and Ukkusuri, S. V. (2026). Understanding Cyber Threats in Cyber-Physical Mobility Systems. In *Advances in Transportation Cybersecurity and Resiliency*. World Scientific.
3. Verma, R., Luo, H., Deodhar, S., **Ka, E.**, Chahine, R., Natu, P., Malhotra, H., Polisetty, V., Thakkar, D. J., Ukkusuri, S. V., Cai, H., Dunlop, S. R., Iyer, A. V., and Gkritza, K. (2023). *Forecasting Shifts in Hoosiers' Travel Demand and Behavior* (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2023/28). Purdue University.

PATENTS AND INTELLECTUAL PROPERTY

1. Ukkusuri, S. V. and **Ka, E.** (2026). System and Method for Trusted Route Guidance with Anomaly Detection. U.S. Provisional Patent Application No. 63/989,950, filed Feb. 24, 2026. Purdue Research Foundation. Patent pending.

Awards, Fellowships & Research Credits

Google Cloud Research Credit Google Cloud	Jun. 2026
Google Cloud Research Credit Google Cloud	Aug. 2025
2025 ITE Great Lakes District Student Paper Award & GLITE Endowment Fund Great Lakes District of Institute of Transportation Engineers (GLITE)	Jun. 2025
Honorable Mention, 2025 Clifford Spiegelman Student Paper Competition The Transportation Statistics Interest Group of the American Statistical Association	Jan. 2025
2025 KOTAA Travel Grant Award Korean Transportation Association in America (KOTAA)	Jan. 2025
Student Paper Award ITE Indiana	Dec. 2024
Kinnier Graduate Scholarship Lyles School of Civil and Construction Engineering, Purdue University	Oct. 2024
STV Civil Engineering Graduate Assistantship Endowment Lyles School of Civil and Construction Engineering, Purdue University	Oct. 2024
2024 KOTAA Travel Grant Award Korean Transportation Association in America (KOTAA)	Jan. 2024
Crooks Travel Scholarship Lyles School of Civil Engineering, Purdue University	Sep. 2023
Outstanding Paper Award 78th Conference of Korean Society of Transportation, Korean Society of Transportation	Mar. 2018
Outstanding Paper Award 74th Conference of Korean Society of Transportation, Korean Society of Transportation	Feb. 2016

Selected Presentations and Invited Talks

1. Physics-Informed AI for Transportation Systems. *CGN 6425: Applied Data Science in Civil and Environmental Engineering*, University of Florida. April 2, 2026.
2. Needs and Gap Assessment of INDOT Workforce Development. *2025 Purdue Road School Transportation Conference and Expo*. March 18, 2025.

3. A Physics-Informed Machine Learning for Generalized Bathtub Model in Large-Scale Urban Networks. *TRB ACP50 Committee on Traffic Flow Theory and Characteristics General Webinar Series*. March 17, 2023.

Academic Service and Professional Activities

ACADEMIC SERVICE

- **Journal Referee:** IEEE Transactions on Intelligent Transportation Systems; Transportation Research Part A: Policy and Practice; Transportation Research Part B: Methodological; Transportation Research Part C: Emerging Technologies; Transportation Research Part F: Traffic Psychology and Behaviour; Transportation Research Interdisciplinary Perspectives; Transportation; Engineering Applications of Artificial Intelligence; Scientific Reports; Electric Power Systems Research; The Journal of Supercomputing; Journal of Cloud Computing; Transportation Research Record: Journal of the Transportation Research Board; Journal of Transportation Engineering, Part A: Systems; Data Science for Transportation
- **Conference Referee:** Transportation Research Board; IEEE Intelligent Transportation Systems Conference (ITSC); ISTTT25 (Co-reviewer); ISTTT26 (Co-reviewer)

PROFESSIONAL ACTIVITIES

- **Association for Computing Machinery (ACM):** Professional Member (Aug. 2025 - Current)
- **American Society of Civil Engineers (ASCE):** Associate Member (May 2025 - Current); Student Member (Sep. 2022 - Apr. 2025); **AI Committee:** Corresponding Member (Aug. 2025 - Current)
- **Institute of Electrical and Electronics Engineers (IEEE):** Member (May 2025 - Current)
- **Institute for Operations Research and the Management Sciences (INFORMS):** Member (May 2023 - Current)
- **Institute of Transportation Engineers (ITE) Purdue Student Chapter:** Communications Director (Aug. 2024 - July 2025); Member (Aug. 2020 - Current)

Technical Skills

- **Transportation Modeling and Simulation :** SUMO, PTV Vissim, TransModeler, TransCAD
- **Geospatial and Statistical Analysis :** ArcGIS, SAS, R, GeoPandas, SPSS, NLOGIT
- **Programming, Machine Learning, and Optimization :** Python, PyTorch, TensorFlow, Keras, PySpark, Java, MATLAB, CPLEX
- **Reproducible Research and Documentation :** Git, Jupyter, LaTeX